

Q6 Smart EV Charging & Parking Management System

Muhammad Usaid 26K-2500

Algorithms

Start

Input Type, Battery, Target, Hours, Time, Member, Disabled, Available.

If Type is neither E nor H, print "error" and stop.

If Battery or Target is outside 0-100, print "error" and stop.

If Hours ≤ 0 , print "error" and stop.

If Time < 0 OR Time > 2359 OR Time MOD 100 ≥ 60 , print "error" and stop.

If Member, Disabled or Available is neither Y nor N, print "error" and stop.

Calculate Needed $\leftarrow$ Target - Battery.

If Needed < 0 , set it to 0.

Set Allowed $\leftarrow 1$, charge $\leftarrow 0$ and Priority $\leftarrow$ "Not Applicable".

Set charging amount, charging discount, parking amount and parking discount to 0.

If Time ≥ 1700 AND Time ≤ 2200 , set Period $\leftarrow$ Peak and Rate $\leftarrow 50$.

Otherwise, set period $\leftarrow$ off-peak and Rate $\leftarrow 35$.

If station is unavailable:

• If Type = "H", set message to "charging unavailable, Parking only".

• If Type = "E", set Allowed $\leftarrow 0$ and message to "No charging slot available".

Otherwise, if Type = "H" AND Battery ≥ 40 , set Allowed $\leftarrow 0$ and does not qualify message.

16. Otherwise, if $\text{Target} \leq \text{Battery}$, set message to "No charging Required".
17. Otherwise, set $\text{charge} \leftarrow 1$ and message to "charging allowed".
18. If $\text{Charge} = 1$.
- If $\text{Battery} \leq 15$ AND $\text{Target} \geq 80$, assign emergency priority.
 - Otherwise, if $\text{Disabled} = "Y"$ OR ($\text{Member} = "Y"$ AND $\text{Battery} \leq 30$), assign priority charging.
 - Otherwise, assign normal charging.
 - Calculate $\text{Charged Amount} \leftarrow \text{Needed} * \text{Rate}$.
 - If Member is Y and priority is not emergency, calculate a 20% charging discount for Off-peak or 10% for peak.
19. If $\text{Allowed} = 1$:
- If $\text{Hours} \leq 2$, set Parking amount to 200.
 - Otherwise, if $\text{Hours} \leq 5$, set it to 400.
 - Otherwise, set it to 700.
 - If disabled status is Y, set parking discount equal to the parking amount.
 - Otherwise, if member is Y, calculate a 20% parking discount.
20. Calculate $\text{Charge cost} \leftarrow \text{Charge amount} - \text{Charged discount}$.
21. Calculate $\text{Park Cost} \leftarrow \text{Parking amount} - \text{Parking discount}$.
22. Calculate $\text{Total discount} \leftarrow \text{Charging discount} + \text{Parking discount}$.
23. Calculate $\text{Final Bill} \leftarrow \text{Charge Cost} + \text{Park cost}$.
24. Print Vehicle type, Battery, Target, Needed charging, Priority, Time period, Both costs, both discounts, Total discounts, final bill and message.

25. If Hour > 8, print "the long stay warning".
26. Otherwise, display "standard parking duration".
27. Stop

Pseudo code.

Begin

Input Type

Input Battery

Input Target

Input Hours

Input Time

Input Member

Input Disabled

Input Available

If (Type != "E" AND Type != "H"), then
Print "Invalid vehicle type".

Input Type.

END IF

If (Battery < 0 OR Battery > 100 OR Target < 0 OR Target > 100)
print "Battery and Target must be between 0 and 100"

Input Battery

Input Target

END IF

IF (Hour ≤ 0), then
 print "Invalid parking duration"
 STOP

END IF

IF (Time < 0 OR Time > 2359 OR Time MOD 100 ≥ 60), then
 print "Invalid time"
 STOP

END IF

IF (Member $\neq$ "Y" AND Member $\neq$ "N"), then
 print "Invalid membership"
 STOP

END IF

IF (Disabled $\neq$ "Y" AND Disabled $\neq$ "N"), then
 print "Invalid Disabled status"
 STOP

END IF

IF (Available $\neq$ "Y" AND Available $\neq$ "N"), then
 print "Invalid station availability"
 STOP

END IF

Needed $\leftarrow$ Target - Battery

If (Needed < 0), then
 Needed $\leftarrow$ 0

END IF

Allowed $\leftarrow$ 1

Charge $\leftarrow$ 0

priority $\leftarrow$ "Not applicable".

Charge Amount $\leftarrow$ 0

Charge Discount $\leftarrow$ 0

Park Amount $\leftarrow$ 0

Park Discount $\leftarrow$ 0

If (Time $\geq$ 1700 AND Time $\leq$ 2200), then

 Period $\leftarrow$ "Peak"

 Rate $\leftarrow$ 50

ELSE

 Period $\leftarrow$ "Off-Peak"

 Rate $\leftarrow$ 35

END IF

If (Available = "N"), then

 If (Type = "H"), then

 print "Charging unavailable, Parking only"

 ELSE

 Allowed $\leftarrow$ 0

 Message $\leftarrow$ "No charging slot available"

 END IF

ELSE IF (Type = "H" AND Battery $\geq$ 40), then
Allowed $\leftarrow$ 0
Message $\leftarrow$ "Vehicle does not qualify for EV charging"

ELSE IF (Target $\leq$ Battery), then
Message $\leftarrow$ "No charging required"

ELSE
Charge $\leftarrow$ 1
Message $\leftarrow$ "Charging allowed"

END IF

IF (Charge = 1), then

IF (Battery $\leq$ 15 AND Target $\geq$ 80), then
Priority $\leftarrow$ "Emergency charging priority"

ELSE IF Disabled = "Y" OR (Member = "Y" AND
Battery $\leq$ 30), then
Priority $\leftarrow$ "Priority charging"

ELSE

Priority $\leftarrow$ "Normal charging"

END IF

Charge Amount $\leftarrow$ Needed * Rate

IF (Member = "Y" AND Priority $\neq$ "Emergency charging priority"),
IF (Period = "Off-Peak"), then

Charge Discount $\leftarrow$ Charge Amount * 20/100

ELSE

Charge Discount $\leftarrow$ Charge Amount * 10/100

END IF

END IF

END IF

IF (Allowed = 1), then

IF (Hours $\leq$ 2), then

ParkAmount $\leftarrow$ 200

ELSE IF (Hours $\leq$ 5), then

ParkAmount $\leftarrow$ 400

ELSE

ParkAmount $\leftarrow$ 700

END IF

IF (Disabled = "Y"), then

ParkDiscount $\leftarrow$ ParkAmount

ELSE IF (Member = "Y"), then

ParkDiscount $\leftarrow$ ParkAmount * 20/100

END IF

END IF

ChargeCost $\leftarrow$ Charge Amount - ChargeDiscount

ParkCost $\leftarrow$ ParkAmount - ParkDiscount

Total Discount $\leftarrow$ ChargeDiscount + ParkDiscount

Final Bill $\leftarrow$ ChargeCost + ParkCost

Print "Vehicle type: ", Type

Print "Current battery percentage: ", Battery

Print "Target battery percentage: ", Target

Print "Charging needed in percentage points: ", Needed

Print "Charging priority: ", Priority

Print "Time Period: ", Period

Print "Charging Discount: ", ChargeDiscount

Print "Charging cost after discount: ", ChargeCost

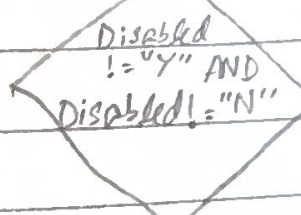
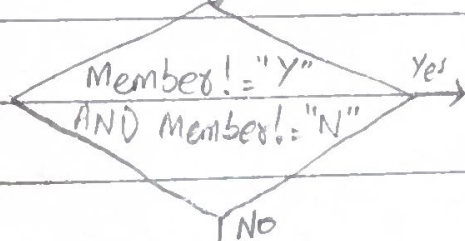
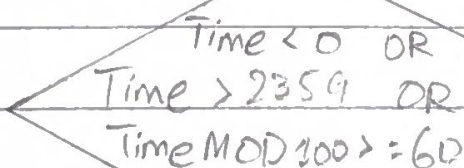
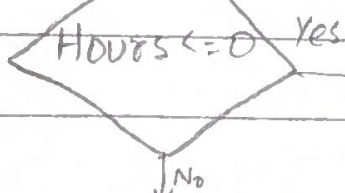
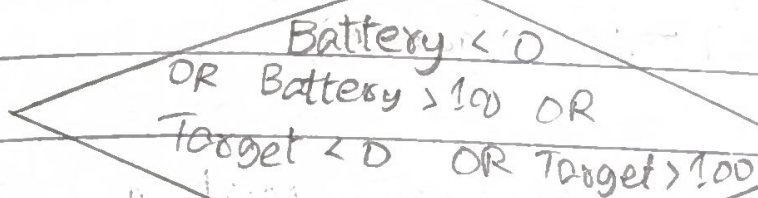
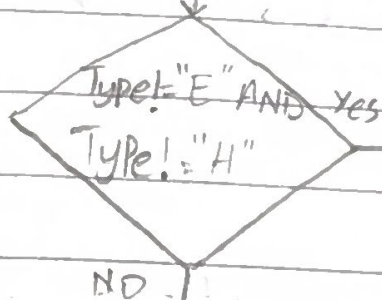
Print "Parking discount:", ParkDiscount
Print "Parking cost after discount:", ParkCost
Print "Total discount:", TotalDiscount
Print "Final Payable amount:", FinalBill
Print Message

If (Hours > 8), then
 Print "Long stay warning: Please relocate your
 vehicle after charging"
ELSE
 print "Standard parking duration"
END IF

STOP

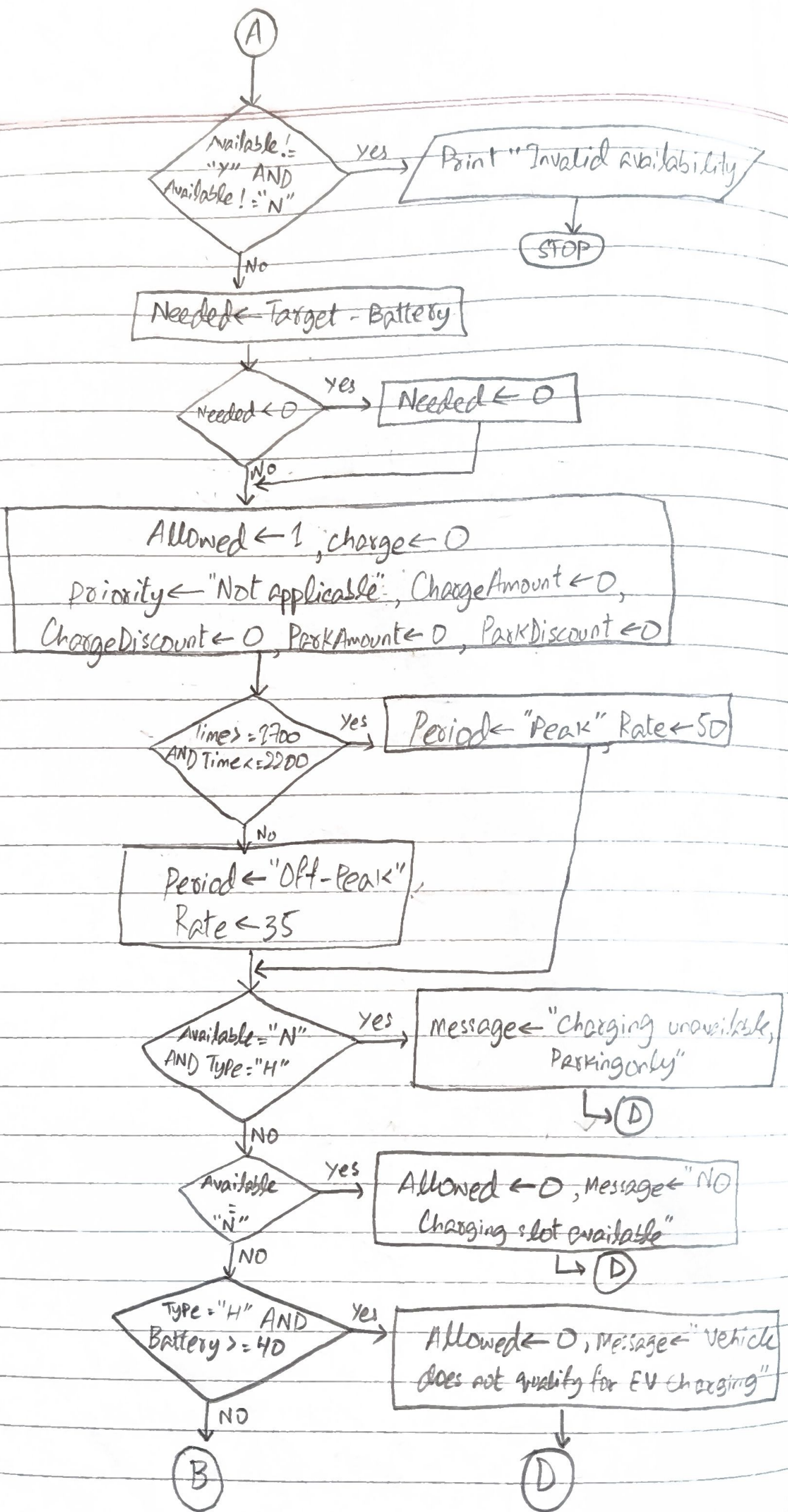
Start

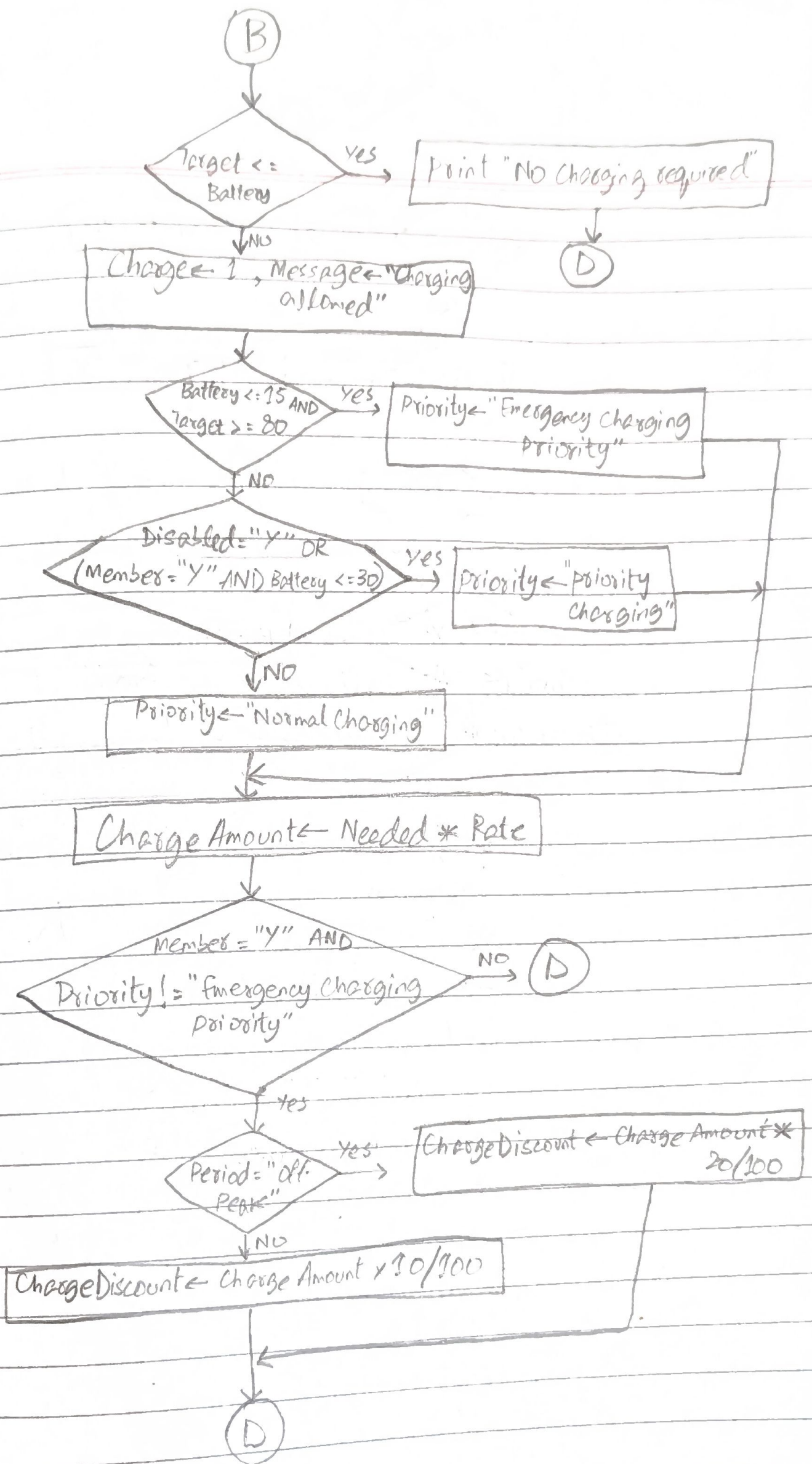
Input Type, Battery, Target, Hours, Time,
Member, Disabled, Available

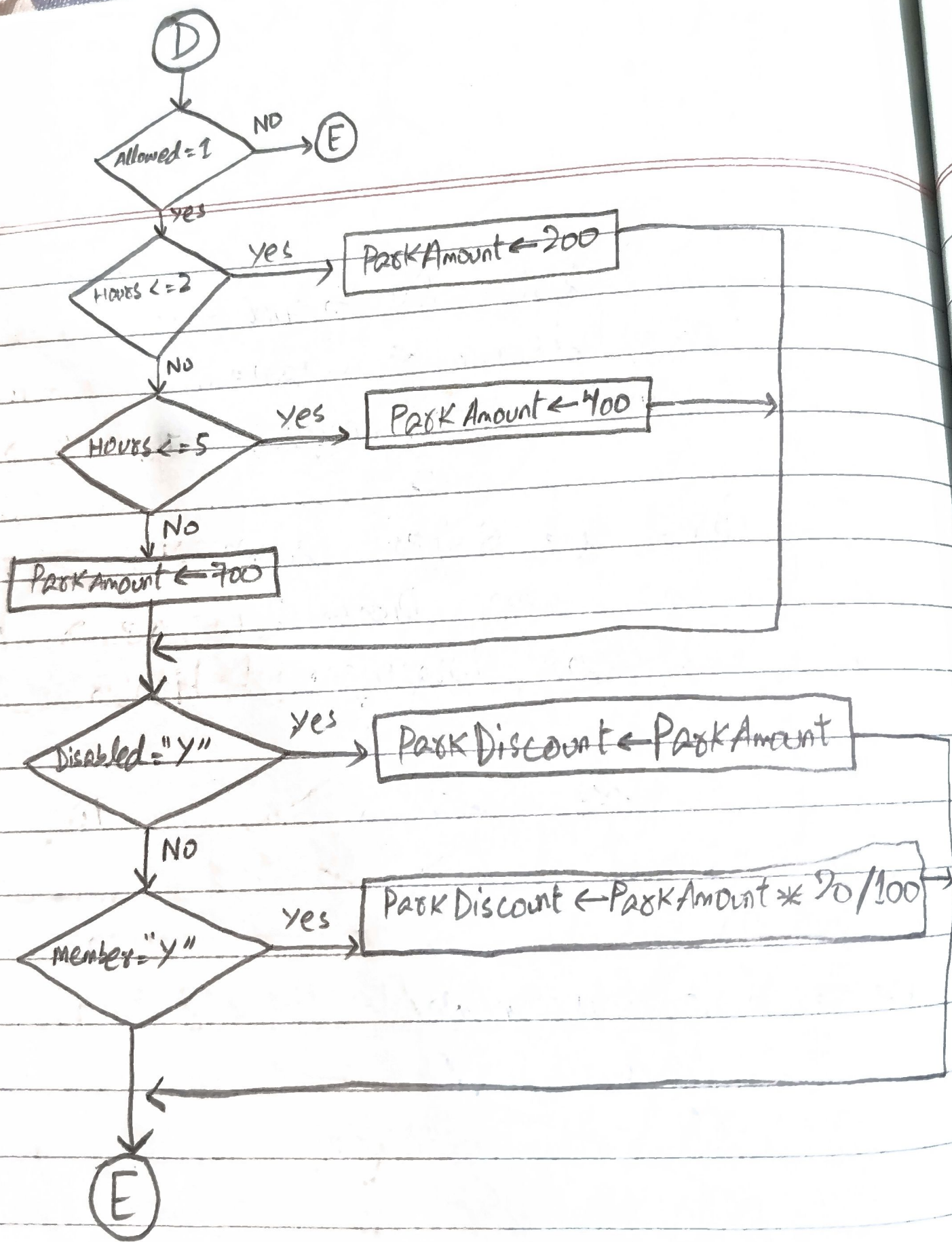


A

STOP







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$\text{ChargeCost} \leftarrow \text{ChargeAmount} - \text{ChargeDiscount}$
 $\text{ParkCost} \leftarrow \text{ParkAmount} - \text{ParkDiscount}$
 $\text{TotalDiscount} \leftarrow \text{ChargeDiscount} + \text{ParkDiscount}$
 $\text{FinalBill} \leftarrow \text{ChargeCost} + \text{ParkCost}$

Print Type, Battery, Target, Needed, Priority,
Period, ChargeDiscount, ChargeCost, ParkDiscount,
ParkCost, TotalDiscount, FinalBill, Message

yes

HOURS > 8

NO

Print "long stay warning:
Please relocate your vehicle
after charging"

Print "standard
parking duration"

STOP